Alexander Tong

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Education

Boston University Expected May 2028

B.S. Mechanical Engineering, Aerospace Concentration GPA: 3.71/4.00

Professional Experience

Boston University Plummer Laboratory June 2026 - Present

Mechanical Engineering Intern *Boston, MA*

- Designed and structurally analyzed a vibration-test fixture in SolidWorks for a 3 kg payload; dynamic FEA predicted 0.44 mm relative deflection at 20.1 Hz with 13 mm peak-to-peak base excitation
- Machined shaker adapter from 3.19 kg stock to 1.90 kg (40.4% reduction) using hand calculations and SolidWorks FEA
- Analyzed bored M10 bolts and mating adapter under preload from 21.7 ft-lbf tightening torque; calculated minimum FOS 2.46 across thread shear, bearing, and pull-through failure modes
- CNC-milled 6061-T6 shaker-fixture on Haas VF-2 Mill to 0.005 in tolerance; generated G-code and engineering drawing
- Fabricated hardware in-house for $71.11 versus a $1,170 outsourcing estimate, reducing direct lab cost by 93.9%
- Designed the test matrix, procedure, and three of four specimen concepts for 48 Instron 68SC-5 tests; processed 0–181 mN force data to investigate physical Ising models

Culturon February 2026 - June 2026

Mechanical Engineering Intern *Sydney, Australia*

- Designed 3 sheet-metal sampling rack variants in Onshape for adhesive-free reactor-port installation, applying bend-relief, flange, and clearance constraints to limit electrical and flow interference
- Planned and executed the company's first chamber-wide nanoparticle deposition study across 4 elevations and magnetic configurations using SEM, optical microscopy, and ImageJ; characterized magnetic polarity-dependent deposition
- Proposed magnetic and fluid-dynamic collection concepts to protect plasma-reactor vacuum pumps

Projects

Aerodynamic Downcomer Fairing August 2026 - Present

Project Lead, Boston University Rocket Propulsion Group

- Leading 4-person team in set-based design of supersonic downcomer fairing, combining empirically-sourced leading-edge, midsection, and boattail geometries across Mach 2-2.81; performed CoDR and trade via first-order drag calculations
- Completed a converged Mach 2.81 Ansys Fluent legacy-fairing baseline, under review for the reuse versus redesign trade

Argo Separation and Recovery Subsystem June 2025 - Present

Co-Lead Engineer, Boston University Rocket Propulsion Group

- Developed hardware in inherited dual-break, dual-deploy recovery (135 lb dry-mass liquid bipropellant rocket; 100k ft)
- Proposed, designed, machined, and integrated a custom CO_2 deployment system after vinyl-charge test failures; optimized mass for Lamé hoop-stress (FOS 2.01) and thread-shear (FOS 6.31) failure; vacuum-tested the assembly
- Designed door, adapter, and seal for a 352 L vacuum chamber; operated deployment tests at 1.1% atmospheric pressure; sized door using Roark plate theory for 1,663 lbf design load and yield FOS 2.12; tested vinyl-charge and CO2 systems
- Dynamically tested 1/2 in Kevlar Brummel splices: estimated 1,800 lbf (1.46x calculated snatch load), no observed slip; proof-load estimate, not ultimate strength; published termination-efficiency comparisons informed selection
- Authored integrated subsystem vinyl black-powder test SOP and led test operations, including crew briefings and fire/abort decisions; supported post-test inspection and failure assessment
- Calculated 823 lbf parachute-opening and 1,230 lbf snatch loads via synthesis of 5 papers; traded hardware accordingly

AIAA Design-Build-Fly Aircraft June 2025 - December 2025

Wing Structural Lead, Boston University

- Led structural, material, and manufacturing trades for two Fusion 360 wing designs; built an XPS/carbon-fiber wing 42.7% lighter than the prior complete wing and cut build time from three months to one

Hybrid Rocket Nozzle September 2024 - November 2024

Lead Engineer, Boston University Rocket Propulsion Group

- Led 5-person team in FEA-driven structural design and optimization of hybrid rocket nozzle for 45k feet ASL, reduced mass 73% with calculated minimum yield FOS 7.8 under mechanical loads; resin-printed and hot-fired for 2 seconds

Technical Skills

SolidWorks, Structural & Dynamic FEA, Ansys Fluent, CNC & Manual Machining, DFMA, Sheet metal, Python, DAQ